



Module 18

Sports Nutrition

Proper nutrition plays a critical role in the health and performance of a runner. Underestimating or misunderstanding proper nutrition and elements of fueling can have disastrous consequences on one's physical performance and health. It is important to understand that while nutritional/fueling guidelines are important to be aware of, individuals will respond differently to the same nutritional/fueling strategy.

❖ OBJECTIVES

Upon completion of this module, you should have an understanding of the following areas:

- Nutritional elements
- Rebound hypoglycemia
- Foot-strike hemolysis
- Hydration strategies
- Dehydration
- Hyponatremia
- Meal timing
- Training and racing strategies
 - Carbohydrate loading theories
- Role of insulin
- Prerace fueling
- Fueling during and after a race
- Training/racing in extreme conditions
- Low-carbohydrate diets
- Weight gain
- What to know about sports supplements
- Popular diets

❖ TERMINOLOGY

Carbohydrate: Organic compound made up of oxygen, hydrogen, and carbon. Categorized as simple and complex, carbohydrates provide energy to the body

Hypoglycemia: Low blood sugar

Hyperglycemia: High blood sugar

Iron: A mineral that is needed to help hemoglobin carry oxygen in red blood cells to the muscles

Electrolytes: Regulate the body's fluids, help to maintain a healthy blood pH balance, and create electrical impulses essential to all aspects of physical activity – from basic cell function to complex neuromuscular interactions needed for athletic performance

Calcium: Element that is important for growth, maintenance, and repair of bone tissue, maintenance of blood calcium levels, regulation of muscle contraction, nerve conduction, and normal blood clotting

Dehydration: Occurs when the body loses excessive amounts of fluids

Hyponatremia: A metabolic condition that occurs when there is not enough sodium (salt) in the body fluids outside the cells. Often associated with overhydration

Carbohydrate loading: Characterized by eating additional carbohydrates in the days leading up to an event with an associated decrease in training volume. The goal of carb loading is to maximize endurance performance by increasing muscle glycogen levels

Insulin: A hormone that is produced in the pancreas and plays a key role in regulating blood glucose (sugar) levels

❖ INTRODUCTION

There is a lot of noise regarding what the right way is to fuel for training and racing. This is likely because nutrition is one of the most important components of training for and participating in a running race. The longer the run, the more important proper nutrition becomes.

Before we go any further, it is important to be clear on one thing: Unless you are a registered dietitian (RD), many states limit the amount of nutritional information you can legally impart to a client. Therefore, in many states you cannot prescribe an eating plan or supplements. What you can do is inform your clients of the purported effects of diets and food items. Put simply, you can educate and inform a client on nutritional information, but you cannot prescribe a nutritional program. If you have clients who are looking for nutritional advice, you should refer them to a reputable RD who specializes in sports nutrition. This is both a legal and performance-based issue.

A 2013 study by Hansen et al. found that marathon runners who followed a scientifically based nutritional strategy were approximately 5 percent faster than runners who self-selected their nutritional strategy (702).

While having a fueling game plan is extremely important regardless of the distance of the race or training run, the longer the run, the more important having a fueling strategy is. The specific types of drinks and foods ingested during a race need to be tried and evaluated during training. This section focuses primarily on the basics of nutrition and nutritional strategies, not selection of food and drink, as one's preference and tolerance is individually based.

It is very common among runners to be so excited and nervous before and during a race that they forget to fuel and hydrate. By developing a fueling strategy and a method to reinforce the strategy, the chance that your client will fuel properly will be greatly increased.

A proper fueling strategy supplies clients with fuel in the right amounts and at the right times so they can finish an event strong and at their desired race pace. Some points to consider are:

- Do clients take drinks/food at aid stops, or do they carry all their own fuel?
 - If stopping at aid stations, which ones to stop at?

- Ideally know ahead of the event what specific items are available at aid stations (e.g., if you know that orange *Gatorade*® will be provided and your client likes this, it is advantageous to train using this specific drink).
- For longer events, advise your clients to set an alarm at regular intervals (15–20 minutes) on their watch to alert them when to consume fuel.
- How often to drink? At what point in the event should the water bottle(s) be empty?

Most of the performance issues that runners experience pertaining to nutrition are due to ingesting too much or too little fuel.

The intake quantities of fuel and nutritional elements that this section suggests are only broad guidelines. The exact intake quantities must be determined on an individual basis. This is most effectively done through consultation with a registered dietician as well as through experimentation during training.

❖ NUTRITIONAL ELEMENTS

FUELING THE ATHLETE

A client's energy needs vary depending on training conditions and intensity. Extreme conditions, such as high altitude, extreme heat, or cold weather, require special dietary considerations. The goal of clients is to determine which foods and fluids they prefer and in what quantities that will meet their energy requirements. A client also needs to learn when to ingest fuel for optimal results.

The *Academy of Nutrition and Dietetics*, *American College of Sports Medicine*, and the *International Society of Sports Medicine* provide recommendations for macronutrient (carbohydrate, protein, fat) intake for endurance athletes. These recommendations provide for optimal nutrition for training, competition, recovery, and overall health in athletes.

Carbohydrates

Carbohydrate intake is a vital part of a running diet, as it provides energy for training and racing. For example, carbohydrates represent 80 to 90 percent of the fuel used to race a 5K and 60 to 70 percent of the fuel required for a half marathon (723). A diet that is between 55 and 65 percent carbohydrates should satisfy the needs of a runner without providing excess that might be stored as fat (412). Carbohydrate intake can be calculated for an individual as $(2.3 - 5.5\text{g/lb/bw/day})$ (722). Carbohydrates maintain blood glucose levels during exercise and replace muscle glycogen. The carbohydrate amount required depends on an athlete's total daily energy expenditure based on *resting metabolic rate* (RMR), training intensity and duration, race distance, gender, and environmental conditions (note: 1 kg = 2.2 lbs).

- Kg = Kilogram
- BW = Body Weight

Regarding carbohydrate drinks (i.e., *Gatorade*), target between four and eight grams of carbohydrate concentration (723).

The type of carbohydrate ingested is just as important as the amount. During carbohydrate-loading periods and when eating meals, it is advised to focus on complex carbohydrates that are found in whole foods and avoid junk carbohydrates such as highly processed foods as they provide little nutrition. Before running and during runs and races, simple carbohydrates are advised as they are easily broken down for immediate energy.

For runs longer than 90 minutes, aim to consume .32 to .45 grams of carbohydrates per pound of body weight per hour.

Several variables increase carbohydrate and caloric requirements:

- **Intensity:** The higher the intensity, the greater the carbohydrate requirement.
- **Heat/Humidity:** Heat and humidity typically increase the carbohydrate requirement.
- **Altitude:** Metabolic rate increases at altitude and will necessitate a higher carbohydrate intake.

Adaptation

Some individuals find that dramatically increasing one's carbohydrate intake causes gastrointestinal issues. If this is the case, gradually introduce higher carbohydrate levels as a program progresses.

Rebound (Reactive) Hypoglycemia

Rebound hypoglycemia refers to a process by which carbohydrates are consumed close to the start of an exercise session and this ingestion causes one's insulin levels to elevate, which in turn causes blood sugar levels to decrease. The net effect of this process is that an athlete feels like he or she has run out of energy, or in endurance athletics terms, "hit the wall."

A 2010 study out of the University of Birmingham analyzed studies that pertained to rebound hypoglycemia and found that while there was a substantial metabolic change in the body (e.g., rebound hypoglycemia), there was not consistent data to point to a decrease in performance (447). This was corroborated by a similar study out of Deakin University (448). However, the University of Birmingham study did find that some subjects experienced effects similar to hypoglycemia, even if they did not have low glucose levels.

It is recommended to experiment with ingesting carbohydrates in varying amounts and at various times prior to exercise bouts to assess the effect on a client (447).

Protein

Protein from both animal and plant sources provides the amino acids needed for muscle growth, repair, and maintenance. An athlete's diet should contain at least 15–20 percent protein, while during peak season some athletes may require up to 30 percent protein, spaced throughout the day (412).

How to calculate an athlete's protein needs:

- **Endurance Athletes:** .6–.8 grams/lb of body weight per day
- **Sedentary Adult:** .4 grams/lb of body weight per day

- **Strength Athlete:** .7–.8 grams/lb of body weight per day

Protein can be a reserve fuel source during an endurance event or training session once glycogen stores have been exhausted. Protein is found in beans, poultry, eggs, dairy, lean meats, and fish. Some energy/sports bars and shakes can supply the appropriate amounts of protein and carbohydrates and are easily digested. While protein is important for building and repairing muscle, it is not a necessary food source right before a race.

PROTEIN SOURCES

Animal: chicken, turkey, fish, milk, yogurt, cottage cheese, eggs, ostrich, lean beef, pork, and protein powder (whey)

Plant: nuts (almonds, walnuts, pecans, pistachios, etc.) seeds (ground flaxseed, chia, pumpkin, hemp), nut butters (peanut butter, almond butter, cashew butter) lentils, beans, minimally processed soy products, edamame, tempeh, nutritional yeast, and plant protein powder (pea, hemp, etc.)

Fat

Fat intake should range from 20 to 35 percent of one's total caloric intake (44–78g: based on 2K calories/day) (723). Fat, which is a source of energy, is important in the diets of athletes. High-fat diets are not recommended for athletes. Heart-healthy fats like those found in fatty fish, avocados, nuts, and olive oil can help with recovery and the inflammation that occurs as a result of exercise. There should be no man-made trans fats (like margarine) in the diet of an athlete (410).

Caffeine

A safe, effective dose of 2.5 mg caffeine/lb body weight (about 375 mg caffeine for 150-pound person) can be consumed in moderation.

Caffeine may make exercise seem easier and enhance performance. Conversely, it may cause nervousness, upset stomach, diarrhea, and jitters. Individuals respond differently to caffeine.

Food	Portion	Caffeine Content (mg)
Coffee, generic brewed	8 oz	133 (range: 102–200)
Starbucks Brewed Coffee (grande)	16 oz	320
Tea, brewed	8 oz	53 (range: 40–120)
Coca-Cola Classic	12 oz	35 (20 oz = 58)
Red Bull	8.3 oz	80
Ben & Jerry's Coffee Ice Cream	8 fl. oz	68
Hershey's Chocolate Bar	1.55 oz	9

Figure 18.1 Caffeine content (404)

A 2008 study from RMIT University in Australia showed that by ingesting carbohydrates in conjunction with caffeine after exhaustive exercise sessions, glycogen resynthesis occurs at a faster rate than with carbohydrates alone (40).

Iron

Iron is needed to help hemoglobin carry oxygen in red blood cells to the muscles. Therefore in regard to endurance athletes, iron is a critical dietary element. While there is typically little difference in vitamin and mineral levels of athletes and nonathletes, athletes are often iron deficient. The recommended daily allowance of iron is 8 mg for men and 18 mg for women. The increased iron requirement for women is due to menstruation.

Following are several common foods and their respective iron content:

Food	Portion	Iron Content (mg)
Tuna	1 can (7 oz)	2
Chicken leg	4 oz	2
Beef	4 oz	3
Spinach	½ cup cooked	1
Raisins	⅓ cup	1
Oatmeal, unenriched	½ cup dry	1
Bran flakes, enriched	1 oz	8

Figure 18.2 Iron content

Foot-strike Hemolysis

Hemolysis refers to damage to red blood cells. There is a substantial amount of research that demonstrates that the impact of running can cause hemolysis (413). Hemolysis after running is most commonly observed in long-distance runners (i.e., marathon) (413, 414).

As hemolysis damages red blood cells, hemoglobin levels are reduced and therefore iron levels are reduced as well (i.e., anemia).

Clients who train for and race in long-distance running events should ensure that they have optimal iron levels.

Electrolytes

Electrolyte is a term used to describe ions. The major electrolytes in the body are:

- Sodium
- Potassium
- Chloride
- Magnesium
- Bicarbonate
- Phosphate

- Calcium
- Sulfate

Electrolytes are minerals that break down into small, electrically charged particles called ions. Present wherever there's water in the body, electrolytes regulate the body's fluids, helping to maintain a healthy blood pH balance and creating the electrical impulses essential to all aspects of physical activity – from basic cell function to complex neuromuscular interactions. The kidneys work to keep the electrolyte concentrations in the blood constant despite changes in the body.

A regular diet provides enough sodium to sustain the body during normal activity levels. However, after three hours of sweating, most people note a performance decline if they are not replacing lost electrolytes. Consequently, many sports drinks contain sodium chloride or potassium chloride.

Some athletes take salt tablets while competing. This should be done only during long-distance events. If taking a salt tablet (200–350mg sodium), take one per hour with six to eight ounces of water.

- 1 lb of sweat = 450–700 mg sodium
- Typical loss in one hour of hard exercise in heat = 800–4,000 mg sodium (403)

Extra salt can be also consumed a day or two ahead of an event. As with carbohydrate loading, the body can store extra salt. This is a big plus for events longer than three hours and/or if it's hot.

Too much electrolyte replacement can cause bloating/water retention and gastrointestinal distress.

Sodium/Electrolyte Replacement Method	Portion	Sodium Content (mg)
<i>PowerAde</i> ®	8 oz	70
<i>Gatorade</i> ®	8 oz	110
Salt	¼ tsp	500

Figure 18.3 Sample sodium contents

Note: A single slice of pizza has approximately 500 mg/sodium.

Calcium

Calcium is important for growth, maintenance and repair of bone tissue, maintenance of blood calcium levels, regulation of muscle contractions, nerve conduction, and normal blood clotting. Inadequate dietary calcium can increase the risk of low bone-mineral density and stress fractures. Female athletes are at greatest risk for low bone-mineral density if:

- Calorie intake is low
- Calcium-rich foods are inadequate or eliminated from a diet
- Menstrual dysfunction is present

Target calcium intake per day is:

Age	Male	Female	Pregnant	Lactating
19–50 years	1,000 mg	1,000 mg	1,000 mg	1,000 mg
51–70 years	1,000 mg	1,200 mg		
71+ years	1,200 mg	1,200 mg		

Figure 18.4 Daily target calcium intake

Sources: Dietary Guidelines for Americans, 2012; Institute of Medicine, 2012

Food sources of calcium include:

Food	Portion	Calcium Content (mg)
Cheese	1 oz	200 mg
Cottage Cheese	.05 cup	65 mg
Soy Milk or Tofu	8 oz	240 mg (varies)
Calcium-Enriched Orange Juice	8 oz	300 mg
Kale, Leafy Green Vegetables	1.5 cups	135 mg

Broccoli, cooked	1 cup	94 mg
Salmon or Sardines with bones	3–4 oz	370 mg
Milk or Yogurt (low fat or fat free)	1 cup	300 mg
Fortified almond milk	8 oz	300 mg

Figure 18.5 Food sources of calcium

Note: Three or four servings are recommended daily.

❖ HYDRATION

Hydration is a heavily debated topic, primarily in the world of athletics. There are two primary schools of thought regarding what the proper hydration strategy and guidelines should be. UESCA will present both strategies/guidelines along with potential shortfalls, as it is likely that the “correct” strategy lies somewhere between the two. Presenting both strategies/guidelines enables coaches to determine what works best for their clients.

Fluid choices depend on the length of activity. For exercise periods of less than 60 minutes, drink water. When training or racing lasts more than 60 minutes, in order to maintain blood glucose levels, it is commonly advised to ingest water plus carbohydrates. A common prescription is to target 100–300 calories per hour from a 6–8 percent carbohydrate solution after the first hour of exercise (411). While carbohydrates in sports drinks can improve performance by maintaining blood glucose levels, it is possible that too many carbohydrates in sports drinks can actually facilitate dehydration. This is discussed below in the section *Dehydration*.

Proper hydration is important for lubrication of joints, thermoregulation of the body, circulation, and to balance water loss.

HYDRATION STRATEGY #1

The established and most adhered to hydration strategy is to drink before feeling thirsty to prevent dehydration (hydration strategy #2). However, research by Dr.

Tim Noakes, a sports science professor at the University of Cape Town, has challenged the norm regarding hydration. Dr. Noakes is the author of *Waterlogged: The Serious Problem of Overhydration in Endurance Sports* (397). In this book, Dr. Noakes looks to debunk what he considers myths regarding hydration in respect to endurance events. His primary point of contention with the accepted school of thought regarding hydration is that there is no preset amount of fluid that needs to be taken in before or during athletics events. Rather, Dr. Noakes states that thirst is the body's self-regulation system to tell it when it needs fluids. In other words, drink only when thirsty.

Below are the points that Dr. Noakes makes regarding hydration based on his studies (399, 402):

1. Mild dehydration does not result in heat exhaustion or other health issues.
 - Mild dehydration does increase the perception of effort.
2. Mild dehydration does not increase core temperature to dangerous levels. An increase in core temperature is most correlated to an increase in intensity (400).
3. Collapsing at the end of endurance events is rarely the result of dehydration or high body temperature (hyperthermia). Rather the cause is typically the result of *exercise-associated postural hypotension (EAPH)* that is caused by a rapid fall in blood pressure. This is most often due to blood pooling in the feet and therefore a lack of blood flow to the brain (398).
4. Hyponatremia is primarily caused by overconsumption of fluids versus an underconsumption of sodium.
5. Drink only when thirsty.
6. Drinking when thirsty is just as effective in controlling sweat loss as drinking set amounts at set intervals.
7. Change in body weight is only one variable when determining sweat loss.
8. Drinking more fluids than what is required to quench thirst does not increase performance.

Of the above points, two (#1, #5) have potential shortfalls.

Point #1

Even if mild dehydration has no impact on one's health, if mild dehydration increases one's perception of effort, this will likely negatively impact performance.

Point #5

Drinking when thirsty versus drinking preset amounts is reliant upon an individual being in tune with the body. For many athletes, the thrill and excitement of competition blurs other aspects of an event. In respect to hydration, oftentimes athletes are so focused on other aspects of the event that they “forget” to drink and eat. This may result in individuals drinking when they are quite thirsty and thus drinking more than they would have when they were initially thirsty. Taking in large amounts of fluids while running may cause stomach cramps, which in turn reduce one’s performance.

Therefore, assuming that drinking when thirsty is the correct strategy, as a coach you must be confident that your client has the focus and body awareness to do so during competition.

HYDRATION STRATEGY #2

As noted previously, this hydration strategy revolves around drinking before feeling thirsty. The thought process behind this is that if you wait to drink until you are thirsty, you are likely already dehydrated to some degree. This strategy typically involves a predetermined hydration strategy, including drinking prior to exercising, with set amounts of fluids being consumed at predetermined intervals during exercise.

FLUID GUIDELINES

When	Fluid Intake Goal
2 hours pre-exercise	16–24 oz (sip, don’t chug!)
30 minutes pre-exercise	8–16 oz*
Hourly during exercising	16 oz /hour
After exercise	Until urine is pale
Daily	Enough to urinate every 2–4 hours

Figure 18.6 Hydration strategy #2: Fluid guidelines

* It is possible that drinking this amount prior to running could have adverse consequences, such as cramping.

HYDRATION CONCLUSIONS

While it is easy to adopt a singular hydration strategy, the correct strategy is likely a hybrid of both the aforementioned strategies, and to Dr. Noakes's point, the right strategy is likely based on an individual versus a preformatted strategy or "rule."

Following is an example of a hybrid hydration strategy:

An athlete prehydrates before a race but once the race starts hydrates only when thirsty.

As trial and error is often the most valuable determinant regarding "correct" training and racing methodologies, it is advised that this approach also be taken in training scenarios regarding hydration, so long as the safety of an athlete is not at risk.

As a point of reference, Dr. Noakes found that when athletes drank when thirsty, they consumed approximately 16–17 fluid oz/hour (397).

DEHYDRATION

Dehydration occurs when the body loses excessive amounts of fluids. The body is then unable to carry out its normal functions and becomes overheated. Severe dehydration can lead to overheating and heat stroke. Consuming fluids during exercise decreases the risk of heat-related illness, maintains physiological function, and improves athletic performance. The symptoms of dehydration may not be obvious until severe dehydration occurs (411). Symptoms of dehydration include dark urine, small volume of urine, elevated heart rate, and headache.

There is debate regarding what percent dehydration causes a decrease in performance and, more important, presents a health risk to an individual. Traditionally, it has been thought that a 2 percent dehydration level negatively affected performance, as figure 18.7 denotes. This determination is based on a paper by the *American College of Sports Medicine* (Sawka et al.) (771). Based

on this guideline, prehydrating before exercising and drinking at set intervals was advised.

Research by Dr. Noakes found that a dehydration level of 2 percent does not necessarily decrease performance. His theory is that a safe level of dehydration is individually based. He notes that for some individuals this may be 1 percent whereas for others it may be 12 percent (397). Regardless of the exact percentage, Dr. Noakes's dehydration theory is that one's thirst response will determine what is appropriate and safe in regard to a hydration strategy.

Percent Dehydration	Physical Effects
1%	Increased body temperature
2%	Impaired performance
5%	G.I. problems, heat exhaustion
7%	Hallucinations
10%	Circulatory collapse

Figure 18.7 Dehydration symptoms (associated with hydration strategy #2)

Urine Color Chart

1	
2	
3	
4	
5	
6	
7	
8	

The urine chart can be useful in determining hydration status. Match color of urine sample to color on chart. Samples that match #1–3 indicate an individual who is well hydrated. Dark urine and strong odor indicate dehydration.

Figure 18.8 Urine color chart

Prevention of dehydration requires an individualized hydration plan and consuming fluid to minimize loss of body weight without overhydrating. Remember to avoid drinking too much or too little fluid, as both can be dangerous. Weigh immediately before and after activity in order to determine average sweat rate. Each pound lost (through sweat) after a workout is equal to 16 ounces of water lost. Knowing a client's sweat rate can help with creating a hydration strategy.

It is likely that a client will sweat more than the individual can replace via fluid intake. This is especially true for long-distance events such as marathons and ultra-marathons.

Determining Sweat Rate

The calculation for sweat rate is as follows:

Pre-weight – post-weight + fluid intake during activity = athlete's individual sweat rate.

Following is an example of this calculation:

- **Pre-workout weight:** 140 lbs
- **Post-workout weight:** 139 lbs
- **Fluid intake:** 16 oz
- **Exercise duration:** 1 hour

$140 - 139 \text{ lbs} = 1 \text{ lb lost (16 oz)} + 16 \text{ oz of fluid consumed} = 32 \text{ oz (2 lbs) of sweat loss/hour}$

Based on two year's worth of data collected by Dr. Noakes, the average sweat range of runners and rowers was 23.7 to 44 fl oz/hr (397). Assuming the average intake of fluids per hour is 16 fluid ounces during long-distance events, an individual will likely become dehydrated, and this equates to a decrease in body weight. The important issue is what level of dehydration will negatively affect performance.

Clothing selection also affects sweat rate. If your client is wearing too many layers, clothes that do not "breathe," or dark colors, sweat rate will be higher than it should be.

Correlation between Carbohydrates and Dehydration

Most sports drinks have a high concentration of carbohydrates. This is often marketed for the purpose of maintaining energy and avoiding hitting the dreaded “wall.” While carbohydrates help to slow the body’s rate of glycogen depletion, is it possible to have too much of a good thing?

According to physiologist and nutrition expert Dr. Stacy Sims, the answer is yes. Dr. Sims notes that when too many carbohydrates are taken in by way of a sports drink, the intestines are forced to take in water from other parts of the body for the purpose of diluting the ingested carbohydrates (760). Drawing away of water from other parts of the body can cause dehydration. Therefore Dr. Sims recommends a lower carbohydrate concentration when drinking fluids that contain carbohydrates – with the sole purpose of the carbohydrates being for fluid absorption – not fuel. As a result, Dr. Sims recommends keeping fluid intake for hydration purposes and food for fueling purposes (671).

HYPONATREMIA

This is a metabolic condition that occurs when there is not enough sodium (salt) in the body fluids outside the cells. Low levels of sodium in the blood can cause cerebral edema (brain swelling). Hyponatremia is primarily caused by excessive drinking behavior, four or more hours of consistent exercise, low body weight, being of the female sex, slow running or pace, race inexperience, high availability of drinking fluids, and extremely hot environmental conditions. Hyponatremia can be life threatening.

Drinking in excess of hourly sweat losses may result in hyponatremia.

Symptoms: bloating, nausea and vomiting, headache, confusion, disorientation, agitation, seizures, respiratory distress, unresponsiveness, coma, and death

In the event of hyponatremia, an athlete should be transferred immediately to a medical facility.

Prevention of hyponatremia can be achieved by determining a client’s hourly sweat rate and the person drinking only enough liquid to match the fluid loss from

sweating. Athletes can add extra sodium to their diet in the days leading up to a race and avoid overdrinking days or hours before a race to prevent lowering blood sodium levels.

Sports drinks containing sodium can help to maintain plasma sodium levels and may reduce the risk of hyponatremia. However, sodium-rich sports drinks will not prevent hyponatremia if an athlete drinks excessively. This is because typically the main cause of hyponatremia is overhydration, not underconsumption of sodium (395).

❖ MEAL TIMING

For the body to digest and convert food into energy, it is important to strategically space all meals and snacks throughout the day. Timing meals and snacks (in addition to creating a hydration strategy) is the first place to begin when creating an eating plan for athletes.

- Following an adequate night's sleep (eight or nine hours), the first meal or snack should be ingested within one hour of waking in the morning.
- If training in the morning, exercise within one to three hours after eating breakfast.
- If training sessions happen in the afternoon, make sure to eat something first thing in the morning and then every three or four hours. Do not eat within one or two hours of training. This will enable adequate digestion and result in better energy during training.
- During training or races lasting more than 60 minutes, the body will need nourishment from easily absorbable sources.
- After a race or training session, the body requires immediate nourishment to replenish muscle glycogen stores, hydrate, and meet protein needs. After this, the athlete can return to regular eating patterns (eating every three or four hours). This is especially the case if a client is engaging in two workouts a day.

- Identifying the best fueling plan while training is critical to developing a race-day fueling/hydration plan. The race-day diet should be as close to the training diet as possible. Do nothing different on race day. This will help an athlete avoid unpleasant digestive issues or dips in blood glucose while competing.
- The key to timing meals and fluids is to train around the same time each day and try to mirror a race schedule. Plan eating times and amounts of food/fluids and test the fueling strategy before, during, and after training.
- Eating a carbohydrate-rich meal three or four hours prior to a training session or race can have positive metabolic effects (blood glucose, carbohydrate oxidation) on the body for at least six hours (448).
- A 2012 study by Res et al. found that casein protein (typically found in milk) ingested 30 minutes prior to sleeping and directly after a strength-training session resulted in effective digestion and a rapid rise in circulating amino acid levels. This resulted in muscle protein synthesis to improve post-exercise recovery (671). Whether or not similar results would be elicited using noncasein proteins will require research in that area.

❖ TRAINING/RACING STRATEGIES

The body can burn upward of 750 or more calories/hour, while the body can digest only 200–250 calories/hour (371).

So what does this mean? It means that runners are essentially a cup with a small hole in the bottom. The goal is not to end up as an empty cup (i.e., hitting the wall). As it is not possible to replace all the calories expended during a race, the goal is to replace enough calories so that the body can continue working without any, or with only a minimal decrease in performance.

It is important to individualize nutritional strategies because each athlete requires different caloric needs and tolerates food and drink differently.

CARBOHYDRATE LOADING

Carbohydrate loading is popular among endurance athletes. *Carb loading*, as it is often called, is characterized by eating additional carbohydrates in the days leading up to an event with an associated decrease in training volume.

The goal of carb loading is to maximize endurance performance by increasing muscles' glycogen levels. The greater the glycogen stores, the longer aerobic activity can last. Through carb loading, one's endurance can be significantly extended, by up to 20 percent (372).



Figure 18.9 Pasta

Archon6812. An Asian-Style "Italian"
Pasta 21 October 2010

Carb loading should be considered when training or racing for 90 minutes or more. Ninety minutes is about the length of time it takes for typical stores of muscle glycogen to begin to run low. Carb loading is more than just eating pasta the night before competition. To fill muscle glycogen stores (carbohydrates stored in the muscle) adequately and in a way the body can handle, athletes need to consume increased levels of carbohydrates several days ahead of an event.

Carbohydrate loading, scientifically termed *glycogen super compensation*, has been around since the late 1960s (276). Since this time, there have been refinements to the practice of carb loading. Following are three main carbohydrate loading theories:

Ahlborg Method

The conventional method was developed by Gunvar Ahlborg, a Swedish physiologist. His method of carb loading was performed over a seven-day period. In Ahlborg's study, an exhaustive workout was performed followed by three or four days of eating primarily protein and fat (10 percent from carbs). The goal of this was to eliminate glycogen stores in the body (glycogen depletion). The next three or four days consisted of ingesting large amounts of carbohydrates (90 percent from carbs) (373). While this approach did increase glycogen stores, it often resulted in fatigue and stress on the digestive system.

No-Depletion Method

In 1981, W.M. Sherman found that the depletion component of the carbohydrate-loading phase might be unnecessary (377). This method reduces the volume of training and increases the carbohydrate intake for three days prior to the competition date. Carbohydrate intake prior to the carbohydrate-loading phase is 55 percent. Once the three-day loading phase begins, carbohydrate intake increases to 70 percent – including on race day (376).

West Australia Carbohydrate-Loading Method

This method could be considered a fusion of the two aforementioned methods as it combines both a depletion and carbohydrate-loading phase. In this 2002 study, seven subjects performed a 150-second effort at 130 percent of their VO_2 max on a bike followed by a 30-second sprint (378). During the 24 hours after the test, the subjects ingested 4.7g carb/lb/bw. This resulted in a 90-percent increase in glycogen levels. This test demonstrated that a substantial increase in glycogen is possible within a 24-hour period.

While this study demonstrated the ability to increase glycogen levels rapidly, it did not have the subjects perform physical activity after the 24-hour period. It is plausible that this method could cause gastrointestinal issues as well as a general feeling of being bloated and full.

There is not a clear consensus regarding the exact amount of carbohydrates that should be ingested during the loading phase. However, 3–5g of carbohydrate per pound/body weight is often suggested. Increases in carbohydrate intake must coincide with a reduction of training volume/intensity.

Carbohydrates ingested during the loading phase should be easily digestible. It is critical to try out a carbohydrate-loading regimen before long-distance-training sessions or minor competitions. This will enable athletes to become familiar and comfortable with the types and quantities of foods and drinks needed to successfully carbohydrate load. Additionally, it will help them get a sense of the performance benefits they can expect as a result. For some individuals, a dramatic increase in fiber can have negative effects on the digestive system. Therefore, any carb-loading strategy *must* be practiced in training to know how the body reacts.

Water is stored with carbohydrates in muscle. Therefore, when the body's muscle glycogen levels are topped off, there is an accompanying increase in water weight (up to 4 lbs). As this weight is solely based on an increase in water due to topped-off glycogen stores, as the glycogen stores lower, so will the water weight (276).

If you feel this added weight will hurt your client's performance during an event, a modified carb-loading process can be implemented in which lower amounts of carbohydrates are ingested. This might result in slightly lower glycogen levels on race day.

The most current research in the area of carbohydrate loading and pacing has been developed by Dr. Benjamin Rapoport (705). Based on his extensive research, Dr. Rapoport developed a calculator to determine the amount of carbohydrate loading required, as well as suggesting pace (751). The link to this calculator is in *Appendix B: Resources*.

What Is Insulin?

As noted in the *Energy Systems* module, insulin is a hormone that is produced in the pancreas and plays a key role in regulating blood glucose (sugar) levels. Individuals who have type 1 diabetes are unable to produce insulin and require insulin shots for the body to uptake glucose from food. The roles of insulin are noted below (449):

- Facilitates muscles, the liver, and fat to uptake glucose from the blood. Glucose stored in the muscles and liver is converted to and stored as glycogen, and glucose stored in fat is stored as triglycerides.
- Prevents the use of fat as an energy source. Therefore when insulin levels are low or nonexistent, the body utilizes fat as a fuel.
- When glucose levels rise, insulin is secreted by the pancreas. When glucose levels fall, the secretion of insulin slows or stops.
- Blood glucose levels rise after ingesting carbohydrates. Simple carbohydrates such as candy bars elicit a greater insulin response (greater secretion of insulin) than complex carbohydrates such as beans. When blood glucose levels rise, this sets off an alarm that signals the

secretion of insulin. This secretion of insulin can cause blood sugar levels to plummet because of the uptake of blood glucose for storage by the muscles, liver, and fat.

- Insulin regulates the metabolism of carbohydrates and fat.

Marathon Performance

Normal carbohydrate (glycogen) stores are insufficient for completing a marathon without hitting the wall (705). Therefore, some form of carbohydrate loading is advised. With the stores of glycogen topped off in the liver, muscles, and blood, it is possible to complete a marathon without ingesting additional carbohydrates midrace (705). While this demonstrates the value of carbohydrate loading, the strategy of relying solely on stored carbohydrates to complete a marathon is not advised.

A 2008 study by Campbell et al. examined the effectiveness of ingesting various forms of carbohydrates midrace (gels, drink, jelly beans) in relation to each other and water (710). The study found that while all carbohydrate sources increased blood glucose over solely water, there was no significant difference in blood glucose levels between the three carbohydrate sources. This study did not discuss individuals' gastrointestinal tolerance to the different types of carbohydrate sources. This factor, as well as one's tolerance to the quantity of carbohydrate intake, is equally important and must be determined during training and on an individual basis.

THE LAST SUPPER

No, we're not referring to a religious event or a death-row meal. We're referring to the last full meal that your client will eat prior to competing in a race. Assuming your client has been carb loading, this is not a "stuff your face" type of meal. Rather, it is a standard-sized meal with approximately 70 percent of calories coming from carbohydrates. Another strategy that some runners employ is to eat frequent small meals. Assuming a race starts early in the morning, the last meal should be no later than 6 p.m. This will allow the body to digest the meal prior to going to sleep.

PRERACE: DAY OF EVENT

The purpose of prerace-day nutrition is to top off liver glycogen stores. This is especially the case if a race starts early in the morning, as the body has used up liver glycogen stores overnight. Muscle glycogen stores remain the same as the night before and represent 80 percent of total glycogen stores. If your client does not replenish glycogen stores the day before a race, there is nothing the individual can do about it on race day. It is not something the body can cram for.

In regard to calories, an individual burns approximately .42 calories/lb/hr during sleep (370). For example, a 150-pound runner who sleeps eight hours will have burned approximately 500 calories during sleep. From the perspective of athletic performance, your client will effectively wake up with a caloric deficit.

Prerace Meal

A prerace meal (three or four hours before a race) should not be substantial, regardless of the distance being raced. For short-distance events, a meal of approximately 200–300 calories would be sufficient, but for long-distance events, a caloric range of 400–900 may be more applicable. If fueling takes place within one or two hours of race time, the caloric intake should be much lower than if fueling three or four hours in advance.

The prerace meal should be easily digestible, a mix of simple/complex carbohydrates, and contain small amounts of protein and fat. Approximately 80 percent of prerace fueling should be carbohydrates (749). A client must also stay hydrated. For clients who can't tolerate eating solid food the morning of a race, smoothies and liquid food replacements are good ideas.

Foods such as bagels, bananas, oatmeal, smoothies, and energy/granola bars typically make for good prerace fueling options. Bananas aside, fruit is often not recommended because of the high fiber content and therefore the increased chance for gastrointestinal issues. All fuel consumed on race day *must* be tried during training to ensure that it can be easily consumed and digested.

Depending on the length of a race and the weight of your client, a prerace meal should consist of 50–75g of carbohydrates. A prerace meal should consist of .5–.9g carbs per lb/bw. Some running and nutrition sources recommend ingesting prerace up to 1.5g lb/bw of carbohydrates before long runs or races such as

marathons. While this may be fine for some runners, many find that they cannot handle this many carbohydrates. Exact prerace caloric and carbohydrate amounts must be determined by an individual's tolerance and determined during the training process.

Best Foods

- Tried and true
- Physiologically comfortable
- Psychologically pleasing

Timing

- Meal: Three or four hours prior to the start
- Small snack: 30–60 minutes prior to the start

For early-morning race starts, some athletes find that setting an alarm to get up during the night to eat is helpful. However, the trade-off is a disruption in sleep that many athletes do not want and, thus, they wake up early to begin fueling.

Start of Race

A client who is hungry can consume a gel 15–20 minutes before the start of a race. Most people require water while consuming a gel. If a client does ingest a gel/snack, the individual must have done it previously in training to ensure it does not lead to an upset stomach or cramps while running. Often, the preferred method of fueling right before the start is a sports drink.

DURING RACE

During a race, it is very easy for clients to get caught up in all the excitement and get thrown off the fueling strategy. This is especially the case with beginners. If your clients carry their own fuel during a run or race, a tip to help stay on track is for them to set an alarm to go off at set intervals to remind them to fuel. This is assuming that they fuel at set intervals versus fueling when thirsty or hungry.

For events lasting longer than one hour, common fuels to maintain glucose levels include:

- Sports drinks

- Diluted juices (1/2 strength)
- Energy bar/gel + water
- Fruit and water

10K or Less

For races 10k or less, it is not necessary to fuel during the event. Assuming a client fueled properly prior to the race, the individual has approximately a 90-minute window in which to race without additional fuel. However, if fuel is used, drinks are typically the best source. The closer your client is to the 90-minute point, the more fueling comes into play.

Half Marathon and Marathon

For marathons and half marathons in which the estimated finish time is greater than 1:30, your client should aim to consume .32 to .45 grams of carbohydrate per pound of body weight per hour. An athlete's tolerance of carbohydrate intake will ultimately determine how many carbs are consumed per hour.

Clients who do not carry their own gels/drinks should be aware of exactly where on the race course the rest stops are and what specific foods/drinks they have. Ideally, this information (e.g., flavor and brand of the gel) can be ascertained prior to the event so that the client can train with these exact fueling products.

Based on your client's weight and food tolerance, the individual should try to take in between 120 and 250 calories/hour when running or racing distances that total longer than 1–1.5 hours in duration.

Remind your client to sip – not chug – water and sport drinks. It is advised to have your client walk through feeding stations while drinking. This enables drinking the whole amount of liquid without spilling it. By walking, the stomach is typically able to process the drink more efficiently than when running. Additionally, water spilled on a client's shoes can increase the chance for blisters later in the run.

If your client is susceptible to stomach cramps when drinking too much fluid before or during a run, it is advised to carry a small bottle (i.e., handheld or flask on belt) while running and sip sporadically throughout an event. Taking smaller

sips and allowing the body to absorb the liquids will reduce the chance of stomach cramping.

If your client is a fast runner and expects to finish a half marathon in 1:30 or less, the fuel requirement during the race will be minimal or none, as the body's glycogen stores can likely support the effort/time.

Fueling for Ultra-marathons

If your client is training for an ultra-marathon, it is strongly advised that the individual consults with an RD who has experience working with this type of athlete, as the nature of the event lends itself to unique and challenging nutritional demands.

AFTER RACE

Eat within 15–30 minutes of finishing. This allows for a two to three times faster uptake of nutrients by the body versus waiting two hours or more to refuel, when the uptake is slower. To optimize recovery, the fuel should be a 4:1 ratio of carbohydrate to protein. This ratio can decrease the post-exercise damage to muscles by 36 percent because of a favorable effect on insulin (379).

Refrain from consuming alcohol for several hours after finishing. Rehydrate! Rehydrate! Rehydrate!

Glycogen stores replenish at 5–7 percent per hour. Therefore, depending on the amount of glycogen stores, full replacement can take upward of 20 hours.

❖ POTENTIAL ISSUES AND EXTREME CONDITIONS

Factoring in training conditions can help an athlete prepare for extreme heat, cold, and changes in altitude. An athlete may benefit from consulting with a *registered dietitian* (RD) for special circumstances where competition is in a

different climate than where training takes place. The athlete will need guidance regarding what to expect and how to adjust fueling and hydration strategies.

Health issues can be prevented or managed with a comprehensive screening before, during, and after training. In addition to an overall health screening, athletes should undergo testing for anemia and macronutrient/vitamin deficiencies. If undetected, these deficiencies can present clinical complications and impair performance. An athlete should have a medical screening and blood work ordered by a physician. If an athlete has an imbalance that can be corrected nutritionally, the individual should consult with an RD who specializes in sports nutrition. The *Academy of Nutrition and Dietetics* has a search engine (www.eatright.org) to help find a registered dietitian.

If traveling overseas or to areas that might not have foods your client is used to eating, it is advised to research the area for suitable food options ahead of time. Your client should also bring or have shipped a supply of favorite and reliable foods.

When training or competing at altitude, both you and your client need to be aware of the following things (379):

- Metabolic rate rises by 30 percent
- 70 percent of altitude-related weight loss is due to muscle loss
- More calories and fluids are needed than at sea level

❖ LOW-CARBOHYDRATE DIETS

As many individuals participate in running events for general fitness and weight loss, not competition, it is likely that many of them participate in some form of low-carbohydrate diet. Three of the more popular low carbohydrate diets are *Atkins*, *South Beach*, and *paleo*.

The paleo diet focuses on foods that are preagriculture (i.e., no grains, dairy, refined sugars, potatoes). Therefore the paleo diet focuses mainly on fish, meat, eggs, vegetables, and nuts (458). Because of a lack of carbohydrates, the paleo diet is typically considered at odds with endurance training. However, some athletes modify the diet to allow for some consumption of grains, but only while training and competing.

Low-carbohydrate diets are also often referred to as *ketogenic diets*. This refers to *ketones*, which are carbon fragments that are formed by the breakdown of fat and provide fuel for the body (176). This section examines low-carbohydrate diets based on their effect on athletic performance, not fat loss or overall health.

ORIGINS AND RESEARCH

Carbohydrates are a necessary element in one's diet as they maintain blood glucose levels during exercise and replace muscle glycogen. Based on many studies, this has been the general consensus over the last century. Nutritional science was developed in the early 20th century and coincided with studies on various diets, including the effects of low, moderate, and high carbohydrate intake (461).

The needle biopsy was developed in the 1960s. This technology allowed researchers to assess intra-muscular fuel stores. This research was the catalyst to the concept of muscle glycogen being a short-term fuel source for high-intensity exercise (462). Additionally, this research also spawned the theory of *carbohydrate loading* (463). Most of the research during this period pointed to fat being of limited use for high-intensity exercise and therefore a low-carbohydrate diet would substantially impair physical performance (461).

Studies also demonstrated that a ketogenic diet could fulfill one's nutritional needs. The MIT and Vermont studies are two of the most cited modern studies regarding the effects on athletic performance in relation to ketogenic-based diets.

MIT Study

The MIT study (1983) by Phinney et al. evaluated five well-trained cyclists after four weeks on a very low-carbohydrate diet (459). The results showed that the vast reduction in carbohydrate intake (<3 percent of diet from carbohydrates) did not negatively affect performance. However, it needs to be noted that the subjects were tested at 62–64 percent of their VO₂ max, likely well below the intensity required during a race. As cyclists were able to perform at this intensity for long periods, a low-carbohydrate diet may increase the muscles ability to become more efficient at burning fat for energy.

In other words, the body becomes more reliant upon fat for energy than muscle glycogen. Subjects in this study realized a decline in performance during the first week of the low-carbohydrate/high-fat diet. However, as the study continued on, the subjects' performances increased with the exception of maximal efforts.

Vermont Study

Like the MIT study, the Vermont study (1980) also demonstrated that athletic performance did not decline during an eight-week inpatient study on a ketogenic diet. (464). Subjects in the Vermont study also showed an initial decline in performance during the first week of the ketonic diet, after which point performance increased. Performance for the Vermont study was assessed as the time to exhaustion during treadmill tests at 75 percent of one's VO_2 max.

Noakes Study

This study had participants divide into two groups: One group consumed a high-fat/low-carbohydrate diet for two weeks followed by three days of carbo loading. The other group consumed a high-carbohydrate diet for two weeks followed by three days of carbo loading (465). The exercise test consisted of cycling for 150 minutes at 70 percent of one's VO_2 max, followed by a 20km time trial.

Members of the high-fat/low-carbohydrate group improved their 20K time by 4.5 percent in relation to the high-carbohydrate group.

CONCLUSIONS

- It is likely that the higher the intensity, the greater the amount of carbohydrate intake is required.
 - The paleo diet allows for an increased carbohydrate intake for those training for high-intensity athletic events.
- Like all other aspects of training, the exact amount of carbohydrate intake will vary on an individual basis.
- All diet modifications and experimentation must be practiced during training to know how the body will respond.
- A zero-carbohydrate diet will likely result in decreased performance.
- If participating in a low-carbohydrate diet, "diet periodization" is recommended for optimal performance.

- Carbohydrate loading prior to an event is recommended.
- A decrease in performance over the first week or two is likely when converting from a normal or high-carbohydrate diet to a low-carbohydrate diet.
 - Symptoms associated with hypoglycemia (low blood sugar) are most likely when initially transitioning to a low-carbohydrate diet.
- High-fat/low-carbohydrate diets seem to be most applicable to low-/mid-intensity exercise bouts.
- During a ketogenic adaptation phase (low carbohydrate), an individual must adhere to the low-carbohydrate regimen. Individuals who occasionally consume carbohydrates during this phase will slow the adaptation phase.
- Assuming an individual is training at high workloads, it is not suggested to adhere to a strict low-carbohydrate diet.

❖ WEIGHT GAIN

Individuals looking to lose or manage their weight often select running as a means to do so. As a result, these individuals often target races ranging from 5K to marathon distance to train for. Conventional wisdom would seem to dictate that the longer the event being trained for, the more applicable it would be for weight loss. Interestingly, this is not always the case.

Weight loss is a generic term that does not relate to fat specifically. *Body composition* is more applicable – as it deciphers between fat and muscle loss/gain. This relates directly to running, as runners often gain lean muscle mass during training. As muscle weighs twice as much as fat, individuals may lose fat but because of an increase in lean muscle mass, their weight may stay the same or increase slightly. Additionally, a runner's body stores water (water retention) as a means to repair damaged muscle fibers and increase glycogen delivery to muscles (670). This increase in water retention may also lead to increased body weight, which is by no means correlated with an increase in body fat.

However, one area that can result in an increase in both fat and body weight is taking in too many calories because of increased hunger. While a runner needs to take in a proper amount of fuel to get through workouts, excessive eating will likely result in weight/fat gain. While increased hunger is normal when training

hard for a running race like a marathon, fuel sources should be nutrient-dense foods that are of high quality. This will likely decrease the chance of eating “empty” calories (food that is not nutrient dense) that don’t serve to fuel the athlete, but rather contribute to weight/fat gain (670). Conversely, eating too few calories while training in an effort to lose weight/fat is not the correct strategy either.

If a runner is looking to lose weight, the rationale of, *“I ran 10 miles today so I deserve to eat this whole cake!”* is not the ideal scenario!

❖ NUTRITIONAL SUPPLEMENTS

If you look at the products being marketed to endurance athletes, you will see many products that claim to assist in the following areas:

- Decrease lactic acid buildup
- Reduce recovery time
- Increase oxygen intake to working muscles
- Increase fat-burning rate
- Replace electrolytes

These are just a few of the areas that sports nutrition markets to endurance athletes. As noted previously, unless you are an RD, it is not advised to tell your client to purchase or use any sports supplement. Just because the FDA does not regulate the supplement industry does not mean that all supplements are either safe or effective. The bottom line is that in regard to sport supplementation, you must refer your client to an RD.

Vegetarian/Vegan

Many positive health benefits can be reaped by following a vegetarian or vegan diet. Runners can consume a plant-based diet and create an effective eating strategy for training and competition. The main areas of focus should be on getting adequate protein, iron, B12, and calcium and to prevent deficiencies. Often this requires some form of supplementation, which as noted previously, should be referred to an RD for advisement.

Caution

In many states, only a registered dietician (RD) can prescribe a nutritional program. Check with your state laws to determine to what extent you can advise your client.

The nutritional information in this certification represents basic guidelines on which you can advise your client. There is a substantial difference between advising your clients on nutritional guidelines and prescribing a nutritional plan. It is your responsibility to operate within your scope of practice at all times!

❖ NUTRITIONAL FACTS

- 70–75 percent of muscle weight is from water
- Low blood sugar releases cortisol, which depresses immunity
- Takes approximately three or four hours for fats in a meal to enter the bloodstream
- Probiotics help enhance digestion and absorption of nutrients
- Eight foods cause 90 percent of all severe food allergic reactions:
 1. Eggs
 2. Fish
 3. Shellfish
 4. Soy
 5. Wheat
 6. Peanuts
 7. Milk
 8. Tree nuts

❖ SUMMARY

- A strategic eating and hydration plan needs to be developed early in the training process.
- Most of the performance issues that runners experience pertaining to nutrition are due to ingesting too much or too little fuel.
- Eat before, during, and after events to speed recovery.
 - Eat within 15–30 minutes of finishing a training session or race.
- Don't over- or underhydrate.
- Carbohydrates maintain blood glucose levels during exercise and replace muscle glycogen.
 - For events or training sessions longer than 90 minutes, aim to consume .32 to .45 grams of carbohydrates per pound of body weight per hour.
- Protein from both animal and plant sources provides the amino acids needed for muscle growth, repair, and maintenance.
- All nutritional strategies and types of food and drink *must* be tried in training.
- Electrolytes regulate the body's fluids, helping to maintain a healthy blood pH balance and creating the electrical impulses essential to all aspects of physical activity.
- Iron is needed to help hemoglobin carry oxygen in red blood cells to the muscles. Therefore in regard to endurance athletes, iron is a critical dietary element.
- Athletes should track their intake and consult with a registered dietitian (who specializes in sports nutrition) for feedback.
- Dr. Noakes states that thirst is the body's self-regulation system to tell it when it needs fluids.
- Hyponatremia is caused when there is an excessively low amount of sodium in the body. This can be caused by overhydration.
- Carbohydrate loading (carb loading) is done in the days leading up to an event for the purpose of increasing glycogen stores.
 - During carb loading, body weight will increase because of an increase in water weight.
- In order for the body to digest and convert food into energy, it is important to strategically space all meals and snacks throughout the day.
- A prerace meal should consist of 1–2g/carbs per kg/body weight.

- During events 10k and less, it is not necessary to eat solid food and the fueling should be on the minimal side.
- Normal glycogen stores are not sufficient for fueling a marathon.
- In many states, only a registered dietician (RD) can prescribe a nutritional program. Check with your state laws to determine to what extent you can advise your clients.
- Unless you are an RD, you are not advised to tell your client to purchase and/or use any sports supplement.
- For medium- to high-intensity training sessions or races, a low-carbohydrate diet is not recommended.